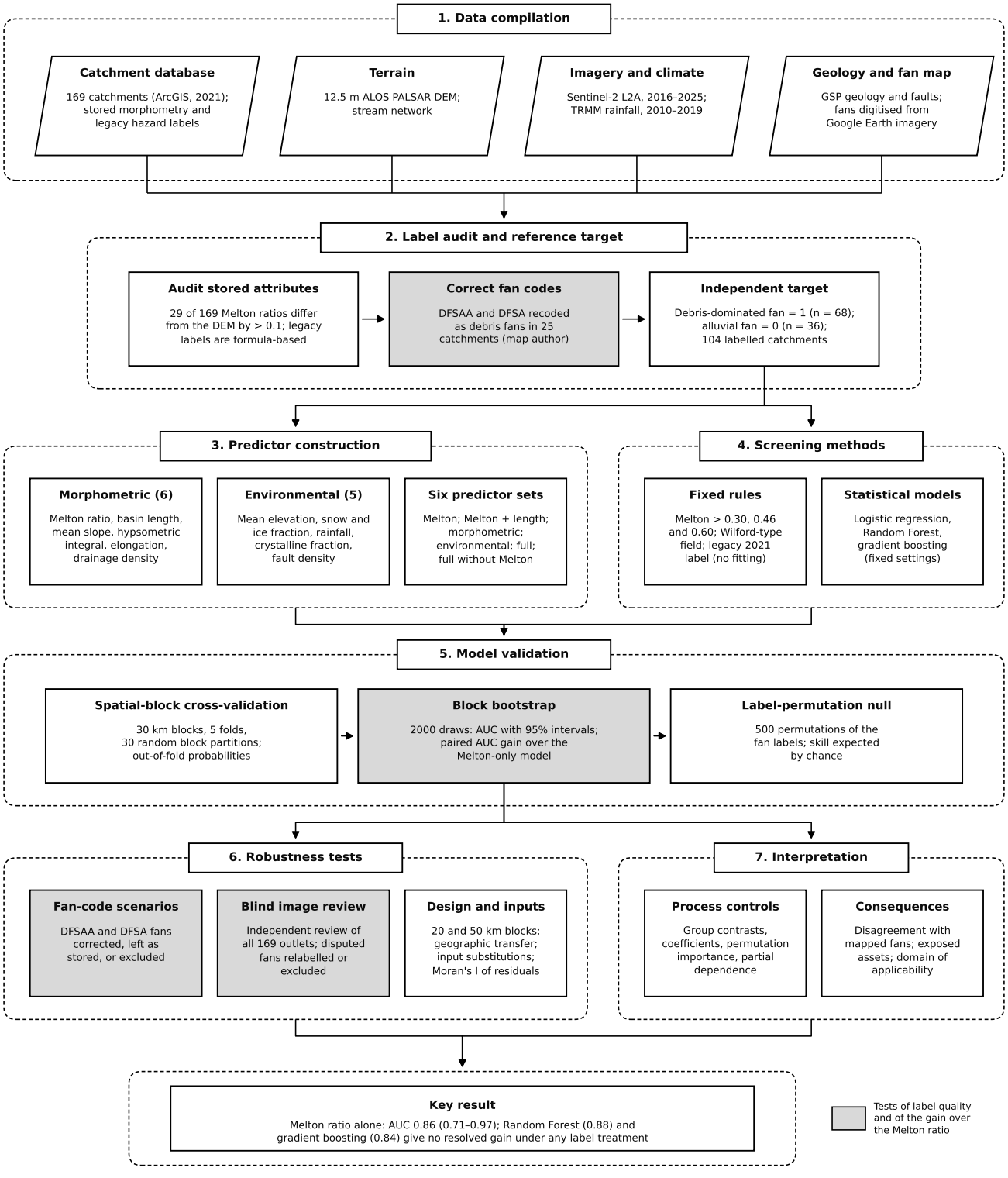
1. Data compilation
Catchment database
169 catchments (ArcGIS, 2021); stored morphometry and legacy hazard labels
Terrain
12.5 m ALOS PALSAR DEM; stream network
Imagery and climate
Sentinel-2 L2A, 2016–2025; TRMM rainfall, 2010–2019
Geology and fan map
GSP geology and faults; fans digitised from Google Earth imagery
2. Label audit and reference target
Audit stored attributes
29 of 169 Melton ratios differ from the DEM by > 0.1; legacy labels are formula-based
Correct fan codes
DFSAA and DFSA recoded as debris fans in 25 catchments (map author)
Independent target
Debris-dominated fan = 1 (n = 68); alluvial fan = 0 (n = 36); 104 labelled catchments
3. Predictor construction
Morphometric (6)
Melton ratio, basin length, mean slope, hypsometric integral, elongation, drainage density
Environmental (5)
Mean elevation, snow and ice fraction, rainfall, crystalline fraction, fault density
Six predictor sets
Melton; Melton + length; morphometric; environmental; full; full without Melton
4. Screening methods
Fixed rules
Melton > 0.30, 0.46 and 0.60; Wilford-type field; legacy 2021 label (no fitting)
Statistical models
Logistic regression, Random Forest, gradient boosting (fixed settings)
5. Model validation
Spatial-block cross-validation
30 km blocks, 5 folds, 30 random block partitions; out-of-fold probabilities
Block bootstrap
2000 draws: AUC with 95% intervals; paired AUC gain over the Melton-only model
Label-permutation null
500 permutations of the fan labels; skill expected by chance
6. Robustness tests
Fan-code scenarios
DFSAA and DFSA fans corrected, left as stored, or excluded
Blind image review
Independent review of all 169 outlets; disputed fans relabelled or excluded
Design and inputs
20 and 50 km blocks; geographic transfer; input substitutions; Moran's I of residuals
7. Interpretation
Process controls
Group contrasts, coefficients, permutation importance, partial dependence
Consequences
Disagreement with mapped fans; exposed assets; domain of applicability
Key result
Melton ratio alone: AUC 0.86 (0.71–0.97); Random Forest (0.88) and gradient boosting (0.84) give no resolved gain under any label treatment
Tests of label quality and of the gain over the Melton ratio